

**INITIALIZE + SELECT**  
800 warm-up steps  
Three separated components



**WRITE HISTORY**  
Two nutrient phases (60 each)  
Then settle 120 steps



**MARK MATERIAL**  
Label each current target mask  
Start bijective tracking



**PREDICTIVE ACCESS**  
Extract fixed scopes  
Leave one entire world out



**CAUSAL BRANCHES**  
Erase / sham / ablate  
Reset N; settle 40; probe 40



**TURNOVER MONITOR**  
Writing remains active  
First all-three  $M \leq 0.25$

Unmet geometry or tracking  
conditions remain in the  
50-world accounting.

Only jointly valid worlds  
enter the reported assays.